Seokweon Jung

Postdoctoral Researcher | Visiting Scholar

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Data Interaction Group, Human-Computer Interaction Institute
Carnegie Mellon University, Pittsburgh, PA, United States

Research Interests

I am a postdoctoral researcher currently serving as a Visiting Scholar with the Data Interaction Group at Carnegie Mellon University's Human-Computer Interaction Institute (HCII). I received my Ph.D. in Computer Science and Engineering from Seoul National University, where I was advised by Prof. Jinwook Seo in the Human-Computer Interaction Lab. My research lies at the intersection of visual analytics, network analysis, and human-centered AI. I develop methods that help people understand complex, evolving information. Particularly, I focus on visual analytics for dynamic graphs that collaborate with AI to adapt information and visual representations to users' contexts. Through cross-domain research collaborations, I aim to build reliable systems that reduce cognitive barriers while preserving human agency.

Experience

- **Carnegie Mellon University** *Mar. 2026 – Present*
  Visiting Scholar (Postdoctoral Researcher), Data Interaction Group, HCII Pittsburgh, PA, United States
- **KAIST** *Sep. 2025 – Feb. 2026*
  Postdoctoral Researcher, LLM Innovation Research Center Daejeon, Republic of Korea

Education

- **Seoul National University** *Mar. 2019 – Aug. 2025*
  Ph.D. in Computer Science & Engineering Seoul, Republic of Korea
  - Ph.D. Thesis: *When Dynamic Graphs Meet Human: Visual Analytics Approach*
  - Advisor: Prof. Jinwook Seo, Human-Computer Interaction Lab
- **Seoul National University** *Mar. 2012 – Feb. 2019*
  B.S. in Computer Science & Engineering Seoul, Republic of Korea

Research Experience

- **SNU-CMU Collaboration Workshop Summer 2024** *Aug. 2024*
  Seoul National University, Republic of Korea
  - Shared knowledge and research experiences in Cognitive, Interactive, and Societal AI with faculty and students of SNU and CMU.
- **Undergraduate Intern** *Sep. 2018 – Feb. 2019*
  Seoul National University, Republic of Korea
  - Developed a hands-free interaction system that recognizes users' head tilts and dispenses spices. The result was submitted and accepted to the 2019 ACM CHI Student Research Competition.
  - Advisor: Prof. Jinwook Seo, Department of Computer Science & Engineering, Seoul National University.
- **D.REAM ACADEMY** *Jun. 2018 – Aug. 2018*
  Yonsei University, Republic of Korea
  - Participated in the 2018 Summer Program for Companion, hosted by Yonsei University's Human and Robot Interaction Center.
  - Designed a chatbot service to help people interact with AI for dementia care.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE FULL, S=CONFERENCE SHORT, W=WORKSHOP/POSTER

Journal Articles

[J.1] **Seokweon Jung**, Jeongmin Rhee, Seoyoung Doh, Hyeon Jeon, Ghulam Jilani Quadri, and Jinwook Seo (2026). **Seeing graphs like humans: Benchmarking computational measures and MLLMs for similarity assessment**. *Information Visualization*, 14738716261434961. DOI: 10.1177/14738716261434961.

[J.2] **Seokweon Jung**, DongHwa Shin, Hyeon Jeon, Kiroong Choe, and Jinwook Seo (2024). **MoNetExplorer: A Visual Analytics System for Analyzing Dynamic Networks With Temporal Network Motifs**. *IEEE Transactions on Visualization and Computer Graphics*, 30(10), pp. 6725–6739. DOI: 10.1109/TVCG.2023.3337396.

[J.3] **Seokweon Jung**, Donghwa Shin, Jinwook Bok, Seokhyeon Park, Hyeon Jeon, Jinwook Seo, Insoo Lee, and Sooyoung Park (2023). **Interactive Visual Analytics System for Criminal Intelligence Analysts with Multiple Coordinated Views**. *Journal of KIISE*, 50(1), pp. 47–56. DOI: 10.5626/JOK.2023.50.1.47.

Conference Full Papers

[C.1] Yungun Kim, Min Hyeong Kim, Suyeon Hwang, Gahyun Kim, **Seokweon Jung**, Hyeon Jeon, and Jinwook Seo (2026). **Democratizing Extensive Reading by Tailoring Web Content**. To be presented at the *ACM Symposium on User Interface Software and Technology* (To appear at UIST 2026).

[C.2] Suyeon Hwang, Minkyu Kweon, Jeongmin Rhee, Soohyun Lee, Seokhyeon Park, **Seokweon Jung**, Hyeon Jeon, and Jinwook Seo (2026). **HookLens: Visual Analytics for Understanding React Hooks Structures**. In *2026 IEEE 19th Pacific Visualization Conference (PacificVis)*, pp. 1–11. DOI: 10.1109/PacificVis68791.2026.00036.

Conference Short Papers

[S.1] Minsuk Chang, Donghun Kim, Hyeon Jeon, **Seokweon Jung**, and Jinwook Seo (2024). **CLeVer: Continual Learning Visualizer for Detecting Task Transition Failure**. In *2024 IEEE 17th Pacific Visualization Conference (PacificVis)*, pp. 369–374. IEEE. DOI: 10.1109/PacificVis60374.2024.00052.

[S.2] Sebeom Park, Soohyun Lee, Youngtaek Kim, Hyeon Jeon, **Seokweon Jung**, Jinwook Bok, and Jinwook Seo (2022). **Vant: A visual analytics system for refining parallel corpora in neural machine translation**. In *2022 IEEE 15th Pacific Visualization Symposium (PacificVis)*, pp. 181–185. IEEE. DOI: 10.1109/PacificVis53943.2022.00029.

[S.3] **Seokweon Jung**, Kiroong Choe, Seokhyeon Park, Hyung-Kwon Ko, Youngtaek Kim, and Jinwook Seo (2021). **Mixed-Initiative Approach to Extract Data from Pictures of Medical Invoice**. In *2021 IEEE 14th Pacific Visualization Symposium (PacificVis)*, pp. 111–115. IEEE. DOI: 10.1109/PacificVis52677.2021.00022.

[S.4] Kiroong Choe, **Seokweon Jung**, Seokhyeon Park, Hwajung Hong, and Jinwook Seo (2021). **Papers101: Supporting the discovery process in the literature review workflow for novice researchers**. In *2021 IEEE 14th Pacific Visualization Symposium (PacificVis)*, pp. 176–180. IEEE. DOI: 10.1109/PacificVis52677.2021.00037.

Workshop Papers and Posters

[W.1] **Seokweon Jung**, Min Hyeong Kim, Hyeon Jeon, and Jinwook Seo (2026). **Readability vs. Faithfulness: Unveiling Correlations between Graph Aesthetics and DR Quality**. In *EuroVis Workshop on GDxDR*. The Eurographics Association. DOI: 10.2312/evgdxdr.20261001.

[W.2] Jiyeon Bae, Mingyu An, Jeongin Park, **Seokweon Jung**, Kiroong Choe, and Jinwook Seo (2026). **Physical Containers as Framing Conditions for Visualization in Augmented Reality**. *19th IEEE Pacific Visualization Conference (PacificVis)*. (Poster) **Best Poster Award**. DOI: 10.48550/arXiv.2603.21637.

[W.3] **Seokweon Jung**, Jiwon Song, Yumin Song, Jinwook Seo, and Ha-kyung Hidy Kong (2026). **A Scoping Review on How HCI Researchers Visualize Results of Thematic Analysis**. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. DOI: 10.1145/3772363.3798541.

[W.4] **Seokweon Jung**, Hyeon Jeon, Jeongmin Rhee, and Jinwook Seo (2025). **Can VLMs Assess Similarity Between Graph Visualizations?** *18th IEEE Pacific Visualization Conference (PacificVis)*. (Poster) **Best Poster Award**.

[W.5] **Seokweon Jung**, DongHwa Shin, Hyeon Jeon, and Jinwook Seo (2024). **Combinational Nonuniform Timeslicing of Dynamic Networks**. *17th IEEE Pacific Visualization Conference (PacificVis)*. (Poster). DOI: 10.48550/arXiv.2404.06021.

[W.6] Kiroong Choe, Seokhyeon Park, **Seokweon Jung**, Hyeok Kim, Ji Won Yang, Hwajung Hong, and Jinwook Seo (2024). **Supporting Novice Researchers to Write Literature Review using Language Models**. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–9.

[W.7] **Seokweon Jung** and Jinwook Seo (2021). **Comparison of node-link diagram and adjacency matrix visualizing network with a group of nodes which have no interconnections**. In *Proceedings of HCI Korea 2021*, pp. 800–803. (Poster).

[W.8] **Seokweon Jung** and Jinwook Seo (2019). **A Visual Analytics Application for Rebalancing Bike-Sharing Systems**. In *Proceedings of KCC 2019*, pp. 1400–1402. (Poster).

Teaching Experience

- **Intelligent Computing System Design Project** *Fall 2024*
  Seoul National University | Teaching Assistant
  ◦ M1522.006600: Intelligent Computing System Design Project (Human-AI Interaction System)

- **Human Computer Interaction** *Spring 2021*
  Seoul National University | Teaching Assistant
  ◦ 4190.426A: Human Computer Interaction

Funded Grants

- **Human Resource Development Program for Industrial Innovation** *Mar. 2026 – Feb. 2027*
  Korea Institute for Advancement of Technology | Researcher
  ◦ Integrating Generative AI into a Dynamic Knowledge Graph Visualization and Analysis Workflow.
  ◦ Grant ID: RS-2024-00436124. Amount: approximately 46,000 USD.

- **Research Grant for Ph.D. Students** *Sep. 2024 – Aug. 2025*
  National Research Foundation of Korea | PI
  ◦ Visualization of Anomaly Detection in Cryptocurrency Transactions with AI and Nonuniform Network Slicing.
  ◦ Grant ID: RS-2024-00410700. Amount: approximately 18,000 USD.

Talks

- **JavaScript-based Open-Source Software for Visual Analytics of Network Data** *Nov. 2025*
  Kwangwoon University, Seoul, Republic of Korea

Honors and Awards

- **Best Poster** *Apr. 2026*
  19th IEEE Pacific Visualization Conference (PacificVis 2026)

- **Best Poster** *Apr. 2025*
  18th IEEE Pacific Visualization Conference (PacificVis 2025)

- **Star Student Researcher** *Mar. 2025*
  Brain Korea 21 (BK21) Four Program

- **Best Visualization Notes** *Apr. 2022*
  15th IEEE Pacific Visualization Symposium (PacificVis 2022)

Academic Services

- **Conference Reviewer (External)** *2025 – 2026*
  IEEE Visualization and Visual Analytics (VIS)

- **Conference Reviewer (External)** *2025 – 2026*
  ACM Conference on Human Factors in Computing Systems (CHI)

- **Conference Reviewer (External)** *2024 – 2026*
  IEEE Pacific Visualization Conference (PacificVis)

- **Student Volunteer** *2024*
  IEEE Visualization and Visual Analytics (VIS)

- **Student Volunteer** *2023*
  IEEE Pacific Visualization Symposium (PacificVis)

REFERENCES

1. **Dr. Jinwook Seo**
   Professor, Department of Computer Science & Engineering, Seoul National University
   Human-Computer Interaction Lab
   Email: jseo@snu.ac.kr
   Relationship: Ph.D. Advisor

2. **Dr. Adam Perer**
   Associate Professor, Human-Computer Interaction Institute, Carnegie Mellon University
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3. **Dr.Ghulam Jilani Quadri**
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4. **Dr. Ha-Kyung Hidy Kong**
   Assistant Professor, School of Information, Rochester Institute of Technology
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